

BULLETS OF FURY

Handoff to Brian - main @ e8f8e62

Mike is handing this over. Everything below is on main. Your drop-0813x is merged - note you pushed to a BRANCH, so it did not reach him until it was merged in. Push to main or open a PR.

YOUR 0813x - MERGED, AND IT SOLVED A BUG I MISSED

The camera-relative spawn offset IS the enemies-appearing-out-of-thin-air report. I investigated that twice and reported stage 1 CLEAN both times, because my probes parked the player at VW/2 or off-screen - the exact positions that hide it. Verified your fix independently: 159px runway at camX 0, 160 and 320, identical. It was 6px to 326px before.

Four of my assertions pinned the old behaviour and failed on merge. I repointed them at your rules - arena floor gated on boss _gen/_mech, and spawn measured against the CAMERA edge at three camX values rather than one.

OPEN - 8 OF 11 ITEMS FROM MIKES LAST LIST

1. Freezer picks up flamethrower but uses ice breath. They are SEPARATE attacks, and ice breath is exclusive to Freezer from level 2 onward.
2. Fire orb and ice orb keep swapping icons, and fire randomly one or the other.
3. fireiceorb fires a basic fireorb.
5. Stage 1 does not move the camera to the blown-up dam after the helicopter boss dies.
6. Stage 2 boss projectiles - Mike calls them awful.
7. Stage came sliding in instead of the lava continuing. Needs a constant-scrolling lava section for the fire boss.
8. Bosses do not die into explosions - they should vanish with the explosion taking over. Only stage 1 does this.
10. Still using plain dialogue where the game has its own boxes. Asked for twice.

Items 1-3 look like one cluster around weapon identity. Item 11 - thruster ANCHOR points matching cinematic to in-game - is also open. Read the flip trap overleaf before touching it.

Traps - each of these cost a session

Every one produced a confident WRONG answer first. Several are mine from the last two days.

Filenames lie. Render a mid-reel frame before you trust one.

ncl_ was used as the CLOUD layer for months - it is chain lightning, gold, rgb(233,186,95). storm800_rc2_master is a storm plate that sat on a SPACE stage. The nhb_boss ICE and LAVA entries are 368x78 HEALTH BAR rigs, not units. A note in the file had already caught the ncl_ case and never fixed the call site, so it shipped anyway.

There are TWO boss spawners with near-identical switches.

spawnSubBoss__inner is minibosses; spawnBoss is stage bosses. Register a kind in the wrong one and you get a 160x120 unit with no name and placeholder art. Tell: spawnBoss on a unit that works in-game fails the same way.

New art MUST be registered in the manifest or it can never decode.

XART.rdy stays false forever and shipBossDraw falls back to a deliberate dark-polygon silhouette. That polygon is a FEATURE - wrong art is recoverable, no art is not. It is the symptom, not the bug.

A diff-based pixel count cannot tell art from a silhouette.

Both are pixels that changed. I passed a placeholder off as working twice on that basis. Distinct-colour count in the unit box separates them: 23k-40k for real art, under 80 for the flat polygon.

THRUSTER_MOUNTS is INLINED in game.js.

assets/data/thruster_mounts.json is a source doc. Editing it alone changes nothing - measured, the JSON read 9/9 while the running game still reported 1/9.

flip in THRUSTER_MOUNTS is an ART-ORIENTATION flag, not a position mirror.

It applies scale(1,-1) about the MOUNT, so the plume extends up into the hull instead of down behind the tail. Only lizzie art is authored for it. Setting it on all nine broke the other eight.

Position is not visibility, and a field read too early is not the field.

Reading e.art on a unit first frame catches a stale placeholder. I reported two bugs off that which did not exist. Diff the frame with and without the unit instead.

Vary the thing the PLAYER controls.

The spawn-offset bug only appears when the player is off-centre. Two investigations with the player parked at VW/2 both came back clean.

The campaign map cannot be rendered in the probe harness.

It is an ASSETS JPEG data-URI that decodes asynchronously, so a probe draws a blank frame. An analysis done on that fallback was worthless. Screenshots are ground truth there.

source-atop clips to the DESTINATION, not the sprite.

That is what drew a rotated dark box around every space prop - the fill landed on the starfield, not the satellite. The cells were innocent throughout.

Ground rules and how to verify

THE SUITE

node _BUILD_SOURCE/test_fl.js - a clean run is 2,659 assertions / 234 sections / 5 failures. Those five are long-standing: preload count, two _superseded ledger checks, volley round count, flash families. MORE than five means check git status for deleted art BEFORE debugging anything - a missing file trips nine assertions across four sections and reads as an unrelated pile of failures.

Zero failures can also mean a crash. Check the assertion COUNT and that the run reaches its final section.

ASSERTIONS THAT FAIL AFTER A DELIBERATE CHANGE

READ them before fixing the code. Several in this file pin a LITERAL LINE rather than the rule - I hit that four times in two days. Rewrite them onto the rule. But do not over-correct: one of my rewrites banned xartTint outright, which would have outlawed the white hit flash that is a standing rule for every unit.

STANDING CREATIVE RULES - MIKE OWNS ALL OF THESE

- Never create placeholder or procedural sprites. Search the existing art first.
- Purple and magenta halos are CONVERTED to a black edge, never deleted.
- Palette and luminance swaps, not overlays.
- Stage biomes are AUTHORED. Stage 1 turning to desert, stage 2 being mostly sand and stage 7 reading as swamp are all intended - do not fix them.
- Stage 7 miniboss stays ratking. Swapping it stopped the sub-boss triggering at all, and the boss gate behind it failed with it.
- Stage 4 boss is ONE plate. Mike: do not split him or seperate him.

A green suite proves state, not pixels. _BUILD_SOURCE/shoot.py and the probe_*.py scripts drive real Chromium and read pixels off the canvas. The render blur that was softening every stage in the game was green in the suite the whole time.